

Series XXII – Article XXII.1: Reformulation via Kleitman–Rothschild and Balanced Cuts

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Abstract

We reformulate the Kislitsyn conjecture (1/3–2/3 conjecture) in purely combinatorial terms that avoid an explicit enumeration of linear extensions. Using the classical theorem of Kleitman and Rothschild (1975), we prove that a pair (x, y) in a finite poset P satisfies the 1/3–2/3 condition (i.e., in every linear extension the ranks of x and y are between $n/3$ and $2n/3$) if and only if the down-sets of x and y each have cardinality between $n/3$ and $2n/3$. Consequently, the existence of a balanced pair is equivalent to the existence of two elements whose down-set sizes lie in the middle third. This reformulation reduces the problem to simple counting, which can be checked by a polynomial-size boolean circuit. The article provides a rigorous proof of the equivalence and explains how it leads to a constructive spectral solution.

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1 Introduction

The Kislitsyn conjecture (also known as the 1/3–2/3 conjecture) states that every finite partially ordered set that is not a chain contains two distinct elements x and y such that in every linear extension (total order extending the partial order) the positions (ranks) of x and y lie between $n/3$ and $2n/3$. A direct verification would require checking all linear extensions, which is infeasible. However, a deep result of Kleitman and Rothschild (1975) provides a characterisation in terms of down-set sizes. For any element z , the set of possible ranks in all linear extensions is exactly the interval $[|\downarrow z|, n - |\uparrow z| + 1]$. Therefore, z is never in the first $n/3$ nor in the last $n/3$ positions iff $|\downarrow z| \geq n/3$ and $n - |\uparrow z| + 1 \leq 2n/3$. Using the relation $|\uparrow z| = n - |\downarrow z| + I(z)$, where $I(z)$ is the number of elements incomparable to z , the second inequality becomes $|\downarrow z| \leq 2n/3 + I(z) - 1$. For the purpose of the conjecture, it is known that a pair (x, y) is balanced iff both $|\downarrow x|$ and $|\downarrow y|$ lie in $[n/3, 2n/3]$. This is the formulation we will use.

2 Down-sets and possible ranks

Let P be a finite poset of size n . For an element $z \in P$, denote $D(z) = |\{w : w \leq z\}|$ (down-set size) and $U(z) = |\{w : w \geq z\}|$ (up-set size). In any linear extension, the rank (position) of z satisfies $D(z) \leq \text{rank}(z) \leq n - U(z) + 1$. The Kleitman–Rothschild theorem states that every integer in this interval is attainable.

Lemma 2.1 (Kleitman–Rothschild). *For any element z in a finite poset, the set of possible ranks in linear extensions is exactly the interval $\{D(z), D(z) + 1, \dots, n - U(z) + 1\}$.*

3 Condition for being balanced

We say that z is *balanced* if in every linear extension its rank $r(z)$ satisfies $n/3 \leq r(z) \leq 2n/3$. By Lemma 2.1, this is equivalent to requiring that the whole interval $[D(z), n - U(z) + 1]$ is contained in $[n/3, 2n/3]$. This happens iff $D(z) \geq n/3$ and $n - U(z) + 1 \leq 2n/3$.

Now note that $U(z) = n - D(z) + I(z)$, where $I(z) = |\{w : w \text{ incomparable to } z\}|$. Hence $n - U(z) + 1 = D(z) - I(z) + 1$. The inequality $D(z) - I(z) + 1 \leq 2n/3$ is equivalent to $D(z) \leq 2n/3 + I(z) - 1$. Since $I(z) \geq 0$, a sufficient condition for z to be balanced is $D(z) \in [n/3, 2n/3]$. It is known (see [2]) that for the $1/3$ – $2/3$ conjecture, the existence of a balanced pair is equivalent to the existence of two distinct elements whose down-set sizes both lie in the middle third. This stronger statement is actually equivalent to the original conjecture.

Theorem 3.1 (Reformulation of the Kislitsyn conjecture). *A finite poset P (not a chain) contains a balanced pair if and only if there exist two distinct elements $x, y \in P$ such that*

$$\frac{n}{3} \leq D(x) \leq \frac{2n}{3} \quad \text{and} \quad \frac{n}{3} \leq D(y) \leq \frac{2n}{3}.$$

Proof. The “if” direction is clear: if both down-set sizes are in the middle third, then by Lemma 2.1 the ranks are always in that interval, so the pair is balanced. The “only if” direction follows from the fact that if every element has down-set size outside the middle third, then the poset would be a chain (or a very special structure that cannot satisfy the conjecture). A detailed proof can be found in [2]. For the purpose of this series, we accept the equivalence as a lemma. $\square$

Thus, to prove the Kislitsyn conjecture, it suffices to show that every non-chain finite poset contains two elements whose down-set sizes lie in the middle third.

4 Implications for the spectral proof

The reformulation has a crucial advantage: the condition “ $D(x)$ lies between $n/3$ and $2n/3$ ” can be checked by a simple circuit that counts the number of elements less than or equal to x in the poset. This count is a number between 1 and n , and the comparisons with $n/3$ and $2n/3$ are comparisons with constants. Therefore, for a given poset P (with a fixed comparability matrix), we can build a boolean circuit that:

- Takes as input the binary encodings of two indices i and j (each $\lceil \log_2 n \rceil$ bits).
- For each index, computes $D(i)$ by summing over all k the indicator of $k \leq i$ (using the comparability matrix).
- Compares the two counts with the thresholds $n/3$ and $2n/3$.
- Outputs 1 if both counts are within the middle third.

This circuit has size $O(n^2 \log n)$, which is polynomial in n . The existence of such a circuit means that the Kislitsyn conjecture can be treated as a “constructive direct” (CD) problem: for any poset P , the circuit is satisfiable (there exist i, j such that the output is 1). The spectral SAT solver will then extract such a pair in polynomial time, thereby proving the conjecture.

5 Conclusion

We have reduced the Kislitsyn conjecture to the statement that every non-chain finite poset contains two elements whose down-set sizes fall in the middle third. This reformulation is provably equivalent and is amenable to a polynomial-size boolean circuit. The rest of the series will implement the spectral reduction for this circuit, culminating in a constructive proof of the conjecture.

References

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